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PRE PSPM
DP024

2024/2025

Instruction: This question paper consists of 7 subjective questions.
Attempt all questions .

QUESTION	MARKS
1	/ 17
2	/ 12
3	/ 9
4	/ 14
5	/ 14
6	/ 14
TOTAL	/ 80

LIST OF FORMULA

1. $\omega = \frac{2\pi}{T} = 2\pi f$

2. $k = \frac{2\pi}{\lambda}$

3. $v = f\lambda$

4. $v = \sqrt{\frac{T}{\mu}}$

5. $y(x,t) = Asin(\omega t \pm kx)$

6. $v = \frac{dy(x,t)}{dt}$

7. $y(x,t) = 2Asin\omega tcoskx$

8. $C = \frac{Q}{V}$

9. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$

10. $C_{eff} = C_1 + C_2 + \cdots + C_n$

11. $\tau = RC$

12. $Q = Q_o e^{-\frac{t}{RC}}$

13. $Q = Q_o\left(1 - e^{-\frac{t}{RC}}\right)$

14. $I = \frac{dQ}{dt}$

15. $Q = ne$

16. $V = IR$

17. $\rho = \frac{RA}{l}$

18. $R_{eff} = R_1 + R_2 + ..R_n$

19. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots \frac{1}{R_n}$

20. $B = \frac{\mu_o I}{2\pi r}$

21. $B = \frac{\mu_o NI}{2R}$

22. $B = \mu_o nI$

23. $\phi_B = BAcos\theta$

24. $\Phi_B = N\phi_B$

25. $\varepsilon = -\frac{d\phi_B}{dt}$

26. $\varepsilon = Blvsin\theta$

27. $\varepsilon = -NA\frac{dB}{dt}$

28. $\varepsilon = -NB\frac{dA}{dt}$

29. $R = 2f$

30. $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

31. $m = -\frac{v}{u}$

32. $m = -\frac{h_i}{h_o}$

33. $y_m = \frac{m\lambda D}{d}$

34 $y_m = \frac{(m+\frac{1}{2})\lambda D}{d}$

35 $\Delta y = \frac{\lambda D}{d}$

LIST OF SELECTED CONSTANT VALUES

Speed of light in vacuum	c	=	$3.00 \times 10^{8}\ \mathrm{m\,s^{-1}}$
Permeability constant	μ_0	=	$4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$
Elementary charge	e	=	$1.60 \times 10^{-19}\,C$
Electron mass	m_e	=	$9.11 \times 10^{-31}\,kg$
Neutron mass	m_n	=	$1.67 \times 10^{-27}\,kg$
Proton mass	m_p	=	$1.67 \times 10^{-27}\,kg$
Deuteron mass	m_d	=	$3.34 \times 10^{-27}\,kg$
Avogadro constant	N_A	=	$6.02 \times 10^{23}\ \mathrm{mol^{-1}}$
Free-fall acceleration	g	=	$9.81\ \mathrm{m\,s^{-2}}$
Density of water	ρ_w	=	$1000 kg\,m^{-3}$ Pa

1(a) A progressive wave is described by the equation $y = 0.28\sin(330\pi t + \frac{5\pi}{4}x)$

where y is in m, x in cm and t is in s.

(i) Determine the amplitude, angular frequency, wave number, wavelength and the direction of the wave propagation.

(ii) Find the displacement of the particle from equilibrium position at t = 17s and x = 29 cm.

(iii) Calculate the vibrational velocity of particle at x = 2.5 cm when t = 3.6 s.

[9 marks]

1(b) A standing wave along a stretched string is represented by the equation

$y = 920cos0.25\pi x sin0.4\pi t$ where x and y are in cm and t in seconds.

(i) Find the distance between two successive nodes.

(ii) Calculate the amplitude of the particle vibrating at x = 500 cm

(iii) Find the displacement of the particle from equilibrium position at t = 23.6 s and x = 27.8 cm.

[6 marks]

1(c) A string that is exactly two metres has a mass of 15 g. A transverse wave travels along this string at a speed of 60 ms^{-1}. Determine

(i) the mass per unit length of the string

(ii) the tension of the string

[2 marks]

2 (a) A silver wire 2.6 mm in diameter transfers a charge of 420 C in 80 minutes.

If this two meters long wire is connected to 14 V, determine

(i) current in the wire,

(ii) resistance of the wire,

(iii) resistivity of the wire.

[4 marks]

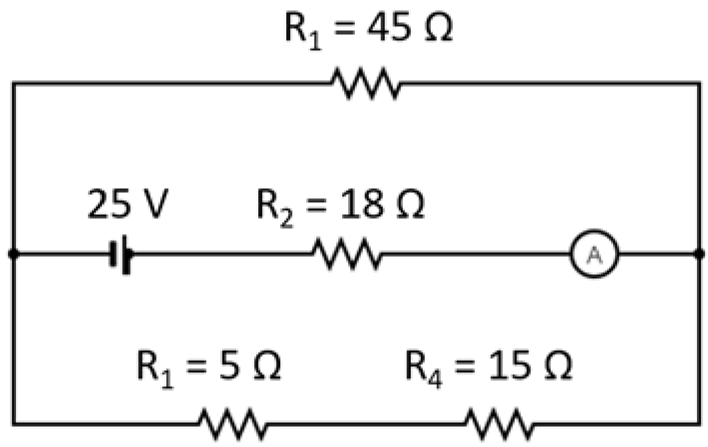

FIGURE 1

2(b) FIGURE 1 shows four resistors connected to a power supply of 25 V.

Determine

(i) the effective resistance of the circuit,

(ii) the ammeter reading,

(iii) the voltage across 45Ω resistor.

[8 marks]

3(a) The potential difference between the plates of a 220 μF capacitor is 5.0 V. Calculate the charge stored on the capacitor.

[2 marks]

3(b)

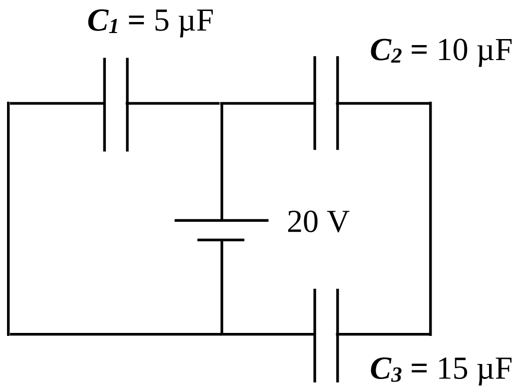

FIGURE 2(a)

FIGURE 2(a) shows three capacitors connected to a 20 V battery. Determine the effective capacitance of the circuit.

[3 marks]

3(c)

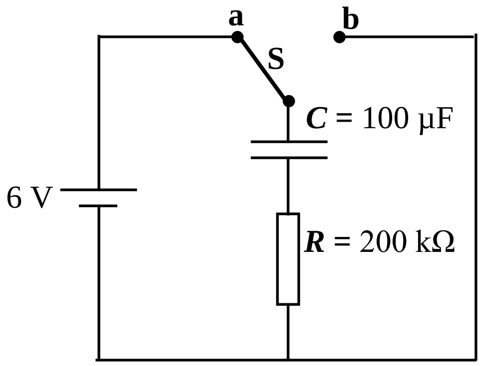

FIGURE 2(b)

FIGURE 2(b) shows a 100 μF capacitor connected in series to a 200 kΩ resistor and a 6 V battery is fully charged. Calculate the

(i) **maximum charge stored in the capacitor.**

(ii) **time constant.**

(iii) **remaining charge in the capacitor at time, t = 5 s when switch S is thrown from a to b.**

[4 marks]

4 (a)

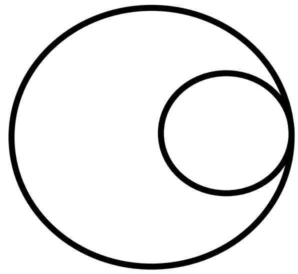

FIGURE 3

Two coil lies on the same plane as shown in the FIGURE 3. The larger coil with the diameter of the 10.0 cm is allowed 5.0 A current to flow in clockwise direction while the smaller coil located within the radius of the bigger coil allowed 3.0 A current to flow in an anti-clockwise direction. Calculate

(i) the magnetic field at the center of the smaller coil and state its direction.
(ii)the resultant magnetic field at the center of the coils when the smaller coil is placed in the middle of the larger coil.

[8 marks]

4(b) A 30 turns coil with diameter of 5.8 cm is placed in a uniform magnetic field of 1.5 T. Calculate the magnetic flux in the coil make an angle of 60˚ to the magnetic field.

[3 marks]

4(c)

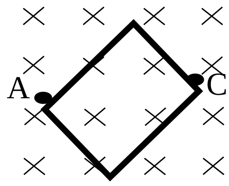

FIGURE 4

FIGURE 4 shows a square coil of 300 turns with a cross sectional area 4 mm^2 is placed perpendicularly to a magnetic field of 7 mT. After 0.3 s, the coil was pulled out between point A and C until the area its reduce to zero.

(i) Calculate the induced emf

(ii) State the direction of induced current.

[3 marks]

5(a) An upright image is produced by a spherical mirror at a distance of 30 cm from an object. The magnification is 4.

(i) State the type of spherical mirror used and explain your answer.

(ii) Calculate object distance.

(iii) What is the focal length of the mirror?

(iv) Sketch and label a ray diagram to show the formation of the image.

[10 marks]

5(b) A camera with a convex lens of 5.0 cm of focal length is used to photograph a girl standing 4.0 m away. The image of the girl is formed at a film behind the lens.

(i) How far must the lens be from the film?

(ii) Describe the characteristics of the image of the girl.

[4 marks]

6(a) In an experiment to determine the wavelength of light using Young's double slit, the following information is obtained:

Slit separation = 0.42 mm,

Distance of screen from slits = 1.6 m,

Fringe separation = 0.26 cm.

(i) What is the wavelength of light?

(ii) How far and in which direction should the screen be moved to change the fringe spacing to 0.32 cm?

(iii) What is the change that can be observed on the screen if the distance between the coherent sources is decreased?

[7 marks]

6(b) A double- slit pattern is observed on a screen 1.4 m away from the slits. If the third order maxima are 29.0 cm apart, determine

(i) the ratio of wavelength and separation between the slits,

(ii) the distance between the fourth order maximum and second order minimum,

(iii) the distance between the fourth minimum on both side,

(iv) the fringe separation.

[7marks]

SOALAN TAMAT